OCP Global Summit 2026: Multi-Dimensional Scalability for AI Network Infrastructures

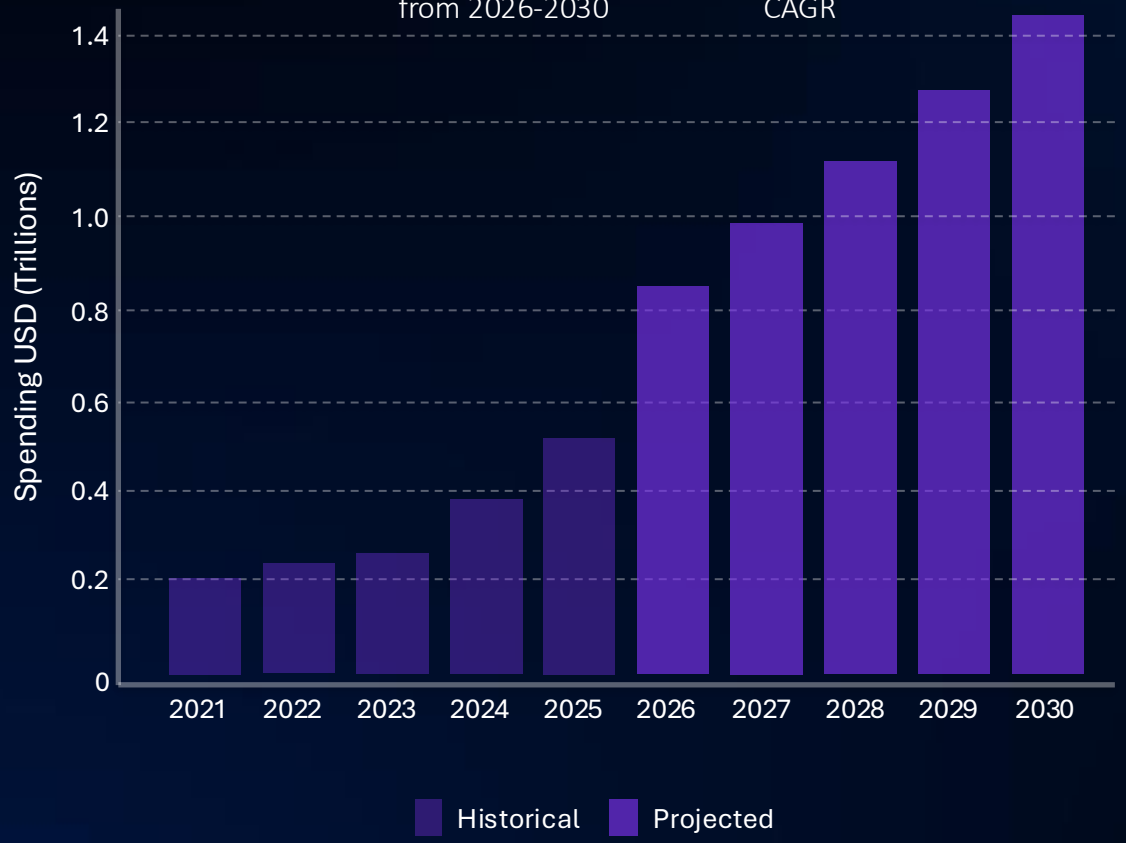
Clayton Wagar
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The AI supercycle

Global data center spending

>$5.5tn
from 2026-2030

23%
CAGR



AI-driven bandwidth growth

23%
Global WAN AI Traffic CAGR
2025-2034

65%
Intra-data center capacity CAGR
2025-2034

Fiber deployments

1.2B Kilometers
2026-2030

Optical interfaces
(ports per year)

	2025		2030
Coherent	970k	→	3m
Intra-DC	50m	→	215m

Source: Dell'Oro, Goldman Sachs

Source: Bell Labs, Corning, Mordor Intelligence, The business Research Company, Light Counting, Cignal AI

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Network infrastructure evolution and the AI supercycle

1. What happened



Generative AI

11,000

Worldwide
data centers

2. What's happening



Agentic AI

>3x

Growth in DCI bandwidth
demands to 3386 EB/month

3. What's coming



Physical AI

21%

At scale AI inferencing
bounded by latency

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AI is the fourth dominant workload

Each wave exceeded the last in volume and became the design driver for a decade

1

Voice

Real-time · small packet · constant rate

Circuit switching, TDM, strict delay budgets

UNIT OF SALE

Minutes

2

Data

Transactional · bursty · delay-tolerant

IP routing, MPLS, QoS, policy control

UNIT OF SALE

Texts, then gigabytes

3

Video

Sustained flows · downstream-heavy

Multicast, CDN interconnect, ECMP, traffic engineering

UNIT OF SALE

Gigabytes

4

AI

Bidirectional · latency-critical · machine-driven

Still being written: this is the conversation

UNIT OF SALE

Tokens

And AI is already large: 1.3 trillion sessions a year and 236 EB of traffic a month. Google alone processes 3.2 quadrillion tokens a month.

All of that is still a human typing into a box and waiting. The machine-to-machine layer has barely started, and each previous wave did not just add volume, it forced a new control plane.

Sources: Nokia Bell Labs; Greg Dorai, SReXperts opening keynote, June 2026; Google I/O, May 2026.

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Distributed AI infrastructure

Device



Distributed



Edge cloud



AI Factory



SMALLER WORKLOADS | LATENCY SENSITIVE | HIGH-BANDWIDTH

PRIVACY & SECURITY | INDEPENDENT OPERATION | DETERMINISTIC BEHAVIOR

PROCESSOR INTENSIVE | COMPLEX CALCULATIONS | MULTI-AGENT COORDINATION

NON-LATENCY SENSITIVE | LARGE DATA MODELS

Healthcare | Autonomous vehicles | Wearables | Retail

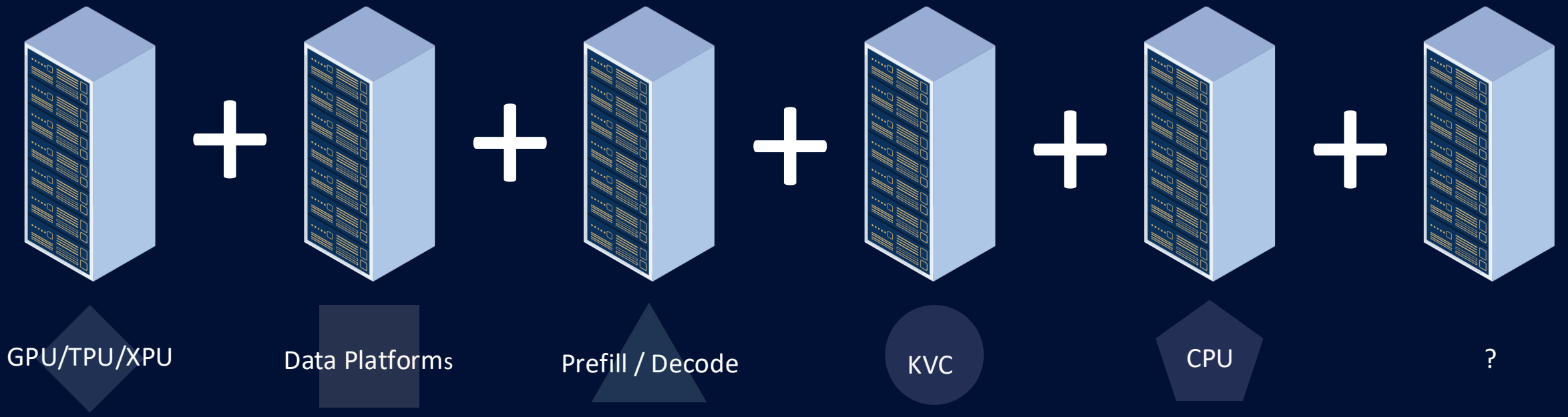
Robots | Drones | Manufacturing | Agriculture

Scientific Simulations | Training | Large scale models

Agents | Video creation

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Disaggregated and Distributed



Deployment increasingly driven by both technology and economic drivers

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Interconnects breaking barriers

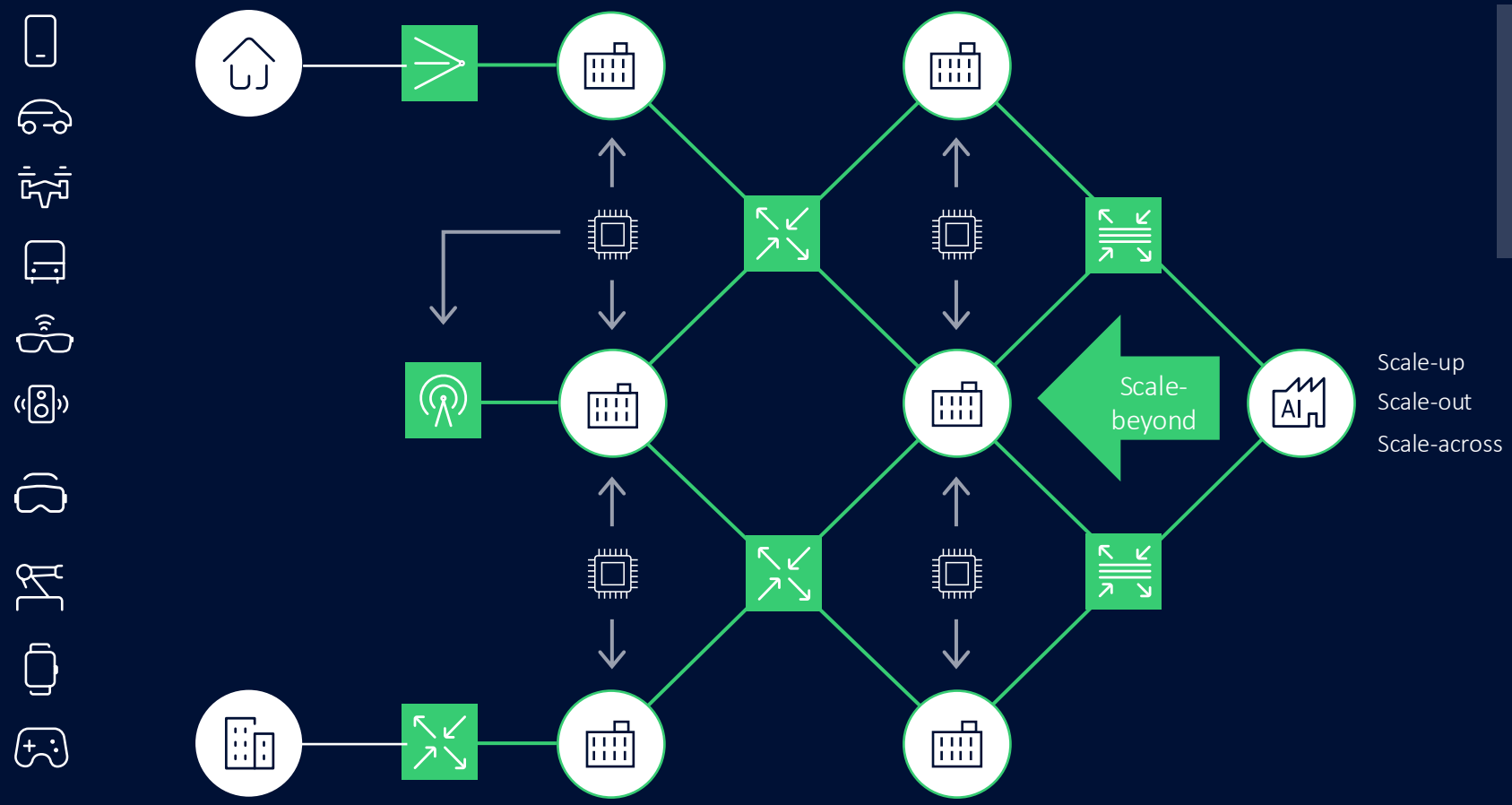
	Scale Up	Scale Out	Scale Across
Purpose	Bigger system	Bigger clusters	Bigger facility
Bottleneck	Compute Density	Cluster Capacity	Building Size

All of these are designed to scale capability within a single operator domain

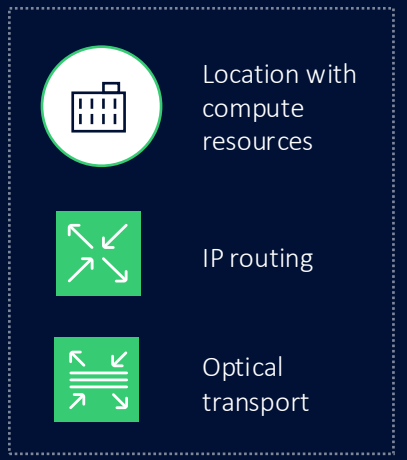
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The Distributed AI grid

Scale-beyond: Advancing connectivity for distributed intelligence



Scale-beyond provides advanced connectivity for distributed AI systems enabling inferencing



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Solutions for AI connectivity

Connectivity Orchestration

- Open interface
- Standards-based
- Multi- Layer

IP routing

- Efficient scalability
- Density
- Capacity expansion
- Deterministic performance
- Deep security

Optical connectivity

- Scalable
- Secure
- Cost, space, & power efficiency
- Open automation

Solutions

Data Center Fabrics / Routing



High-performance platforms
- 800 Gb/s coherent routing
- High density 800GE ports
- Ultra low power consumption



AI-powered automation
- K8s Intent-based policy platform
- Agentic AI human interfaces
- Closed loop automation



Security and trust
- Quantum-safe networking
- DDoS mitigation
- IP Trusted Systems

Transceivers

Coherent Pluggables
Cost, space, Power Optimized





Full band transponders
Operationally efficient scalability

Line Systems



Modular Solutions



Expanded spectrum
11.1Ghz Super-C + Super-L



Direct attach
Hyperscale OLS



Quantum Security
Encryption,
Certified key management, distribution, assurance

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Four tiers, one metric — and only one fell behind

Compute Intensity Ratio (CIR) — the bytes a tier can move for every FLOP the accelerator can do, in millibytes per FLOP

TIER	INTERCONNECT	CIR (mB/F)	vs NVLINK
Memory	HBM	3.39	3.7× above
Scale-up	NVLink	0.910	baseline
Scale-out	InfiniBand NDR / 400G Ethernet	0.006 – 0.050	18 – 152× below
Beyond	**Conventional WAN (1–10 GbE)**	**0.0000158 – 0.000158**	**5,760 – 57,600× below**

Four to five orders of magnitude open up at the last tier — and that is the tier every distributed workload has to cross.

And the shape of the gap is the argument: the tiers the industry funded have broadly tracked compute — the B200 generation holds on-package CIR at ~3.56 mB/F against H100's 3.39. The wide-area tier is the one that has not, and it widens 12–19% a year.

H100 at 989 TFLOPS dense BF16; all compute figures dense. Ranges reflect link class and oversubscription. Method and derivation: arXiv:2608.14967.

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LIGHTSTORM

Training tolerates the WAN. Inference does not.

The same link, two workloads — and the honest bound on how large the inference number really is

TRAINING

5 Gbps

Gradient exchange is periodic and latency-tolerant. The WAN absorbs it. An assumed rate, not a measured one.

INFERENCE

2,147 Gbps

10.74 GB of KV cache moved inside a 40 ms time-to-first-token budget. 70B, multi-head attention, 4K context.

429× more wide-area demand from inference than from training — and inference is the workload that is scaling.

But 2,147 Gbps is the multi-head-attention bound. Grouped-query attention at 1/8 KV heads needs 268 Gbps; latent attention about 58 Gbps. The claim is that cross-site state transfer stops being exotic — not that everyone needs heroic pipes.

Modelled; propagation excluded. Derivation and the full attention table: arXiv:2608.14967 §1–2.

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So we are going to measure it — in the open

Ten metrics on a new deep-buried production corridor in India: three sites, multi-vendor open line systems under interoperability test, real RDMA-class traffic over real wavelengths

1

THE OPERATOR

Provisioning and capacity-step time, delay-variation and protection percentiles, multi-tenant isolation, unit cost. Only an operator can measure these.

2

THE VENDORS

Optical line systems and packet fabrics in interoperability test — transport behaviour under RDMA-class traffic, measured rather than asserted.

3

THE INFERENCE ECOSYSTEM

Measured TTFT and GPU utilisation, the real transfer-versus-recompute flip point, and RDMA running over real wavelengths — without which the transport numbers stay analytic.

The pilot window opens late 2026 and is deliberately open to industry collaborators. The method and the results are published either way — we would rather run it with you than describe it to you.

The paper, the model and the ten-metric plan: arXiv:2608.14967 · arxiv.org/abs/2608.14967 · Prasanna C, Lightstorm

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When Does Distributed AI Inference Need More Wide-Area Bandwidth?

A Co-Design Evaluation of Optical, Packet, and Software Levers

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August 2026

arXiv:2608.14967v1 [cs.DC] 15 Aug 2026

Abstract

Wide-area bandwidth per unit of GPU compute falls every hardware generation: in compute-intensity-ratio terms (CIR, bytes per FLOP), the gap between on-package memory and the conventional WAN is four to five orders of magnitude, widening at roughly 12–19% per year. Position papers—including our own earlier work—argued that this makes elastic optical wide-area capacity necessary for cross-site AI inference. Reviewers correctly objected that such arguments show more bandwidth *helps*, not that provisioned bandwidth beats the alternatives: KV recomputation, cache compression, locality-aware routing, scheduling, or an overprovisioned packet backbone. This paper does the comparison. We derive a workload model predicting when moving inference state across sites beats recomputing it—a context-independent crossover at 74–111 Gbps per stream for a 70B multi-head-attention model, falling to 9–14 Gbps under grouped-query attention at 1/8 KV heads—and quantify five sensitivity axes: context length, attention architecture, concurrency (queueing), agentic multi-step compounding, and loss/jitter-induced effective-bandwidth collapse. We bound the economics: at list GPU prices recomputation is cheaper than transfer; transfer wins when GPU scarcity and KV reuse multiply the effective GPU cost by roughly 5–20×, and modern attention moves the breakeven an order of magnitude in transfer's favour. We position the network levers correctly rather than adversarially: packet networks allocate lit capacity at millisecond timescales; optical fungibility changes the amount of lit capacity at minute timescales, substituting for packet overprovisioning economically rather than functionally. Finally, we specify a ten-metric measurement plan on a three-site production-fibre testbed and frame it as an open ecosystem exercise: no single company can—or should—assemble the end-to-end evidence alone. Every claim is bounded by the regime in which it holds; several findings weaken the naive version of our own thesis, and we state them.

About this paper

This paper consolidates an operator's interconnect analysis after a full review cycle: two academic submissions in 2026 whose seven reviewers validated the core claim while correctly demanding that it be proven against software alternatives rather than argued. Everything analytical is backed by a reproducible model with falsifiable predictions; a ten-metric measurement plan is defined on a new deep-buried production fibre corridor, with a pilot window in late 2026 deliberately open to industry collaborators. The frame is the GPU-to-GPU continuum—accelerator → HBM → NVLink/scale-up → scale-out fabric → scale-across WAN → storage—in which the industry has invested at every tier except one. This paper supplies the operator's quantified case for that missing tier: when it binds, when it does not, and what it must guarantee.

https://arxiv.org/abs/2608.14967

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